George Panagopoulos

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Experience

- **University of Luxembourg**, Belval, Luxembourg (01.2024 – Current)
  Postdoctoral Scientist (01.2024 – 01.2026) → Research Scientist (PI) (02.2026– Current)
  - Graph neural networks for **causal effect estimation**.
  - **Active learning** and adaptive experimental design.
  - **Gene perturbation modeling** for transcriptomic experiments.
  - **Secured competitive research funding**: 499k (FNR CORE Junior 2025), 331k (FNR HPCAI BRIDGES 2025).
- **Amazon**, Applied Scientist, Luxembourg, Luxembourg (04.2022 - 06.2023)
  - Machine learning for routing (internship 4-10/2022), operations research (FTE 12/2022-6/2023).
- **Ecole Polytechnique Informatics Laboratory (LIX)**, Research Engineer, Palaiseau, France (06.2018 - 03.2022)
  - **Graph learning** for forecasting and combinatorial optimization.
  - **Data mining** from bibliographical networks, medical ontologies and legal documents.
- **University of Houston, Computational Physiology Lab**, RA/TA, Houston, TX (08.2016 - 05.2018)
  - **Forecasting** driving anomalies using physiological signals from wearable sensors.
  - **Statistical analysis and inference** on physiological data.
- **National Center for Scientific Research Demokritos**, Research Assistant, Athens, Greece (08.2014 - 07.2016)
  - **Brain computer interface** experiments and tutorials for university and high-school students.
  - **Natural language processing** modeling computational creativity in multi-lingual essays.

Education

- **Ph.D. in Computer Science**, Ecole Polytechnique, Paris, France (01.2019 - 03.2022)
- **Ms. in Computer Science**, University of Houston, Houston, TX (08.2016 - 05.2018)
- **Bsc. in Informatics and Telematics**, Harokopio University of Athens, Greece (09.2010 - 07.2014)

Selected Publications

1. G. Panagopoulos, J.F. Lutzeyer, S. Ennadir, M. Vazirgiannis, J. Pang *"Efficient Data Selection for Training Genomic Perturbation Models"* in arXiv, 2025
2. R. Bresson, G. Nikolentzos, G. Panagopoulos, M. Chatzianastasis, J. Pang, M. Vazirgiannis. *"Kagnns: Kolmogorov-arnold networks meet graph learning"* to appear in Transaction on Machine Learning Research (**TMLR**), 2025
3. G. Panagopoulos, D. Malitesta, F. Malliaros, J. Pang. *"Uplift Modeling Under Limited Supervision"* European Conference on Machine Learning and Knowledge Discovery in Databases(**ECML PKDD**), 2024
4. G. Panagopoulos, N. Tziortziotis, M. Vazirgiannis, F. Malliaros. *"Maximizing influence with graph neural networks"* Advances in Social Networks Analysis and Mining (**ASONAM**), 2023
5. B. Rozemberczki , P. Scherer, Y. He, G. Panagopoulos, M. Astefanoaei, O. Kiss, F. Beres, N. Collignon, R. Sarkar. *"PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models"*. in Conference on Information and Knowledge Management (**CIKM resource track**), 2021
6. G. Panagopoulos, G. Nikolentzos, M. Vazirgiannis.*"Transfer Graph Neural Networks for Pandemic Forecasting"*, in AAAI Conference on Artificial intelligence (**AAAI**), 2021
7. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Multi-task Learning for Influence Estimation and Maximization"*, in **IEEE Transactions of Knowledge and Data Engineering**, 2020, 10.1109/TKDE.2020.3040028
8. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Influence Maximization using Influence and Susceptibility Embeddings"*, in AAAI International Conference on Web and Social Media (**ICWSM**), 2020
9. G. Panagopoulos, I. Pavlidis. *"Forecasting Markers of Habitual Driving Behaviors Associated with Crash Risk"* **IEEE Transactions of Intelligent Transportation Systems**, 2019
10. G. Panagopoulos. *"Multi-Task Learning for Commercial Brain Computer Interfaces"*, in Proc. of IEEE BioInformatics and BioEngineering (**BIBE**), Washington DC, USA, October 23-25, 2017

Technical Skills

- Coding: Python (advanced), Java (intermediate), Scala, C++ (basic)
- Machine Learning: PyTorch, PyG (advanced), JAX, Tensorflow (intermediate), PyMC (basic)
- Statistics & optimization: R (advanced), Matlab, XPRESS (intermediate), Julia, CVXPY (basic)
- Databases: MySQL, PostgreSQL (advanced), AWS ETL (intermediate), Spark, MongoDB, Hadoop (basic)